

PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 60 /Pass Marks: 24

MCA104/MCA114: Advanced Database Management System (New/Old)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. Check whether the schedule is conflict serializable or not. Also find out equivalent serial schedule.

S: R2(A); R1(B); W2(A); R3(A); W2(B); W3(A); R2(B); W2(B)

2. Consider the schema for Company database:

EMPLOYEE(SSN, Name, Address, Sex, Salary, SuperSSN, DNO)

DEPARTMENT(DNo, DName, MgrSSN, MgrStartDate)

DLOCATION(DNo, DLoc)

PROJECT(PNo, PName, PLocation, DNo)

WORKS_ON(SSN, PNo, Hours)

Write SQL queries to

- (a) Make a list of all project numbers for projects that involve an employee whose last name is 'Scott', either as a worker or as a manager of the department that controls the project. 2
- (b) Show the resulting salaries if every employee working on the 'IoT' project is given a 10 percent raise. 2
- (c) Find the sum of the salaries of all employees of the 'Accounts' department, as well as the maximum salary, the minimum salary, and the average salary in this department. 4
- (d) Retrieve the name of each employee who works on all the projects controlled by department number 5 (use NOT EXISTS operator). 2
- (e) For each department that has more than five employees, retrieve the department number and the number of its employees who are making more than Rs. 6,00,000. 2

Contd. ...